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Contents Table

Functions

- Vertical Line Test
- $|x|$
 - Basic Definition of Absolute Value: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$
 - $\sqrt{x^2} = |x|$
 - $e^{\frac{-|x|}{2}}$ (graph this)
- Anton – Chapter 0 – Page 8: “The Effect of Algebraic Operations on the Domain”

Limits

1. Easy ones

- **Direct Substitution**

Direct Substitution – Theorem 1.4.10

If $f(x)$ is a polynomial or rational function, and a is in the domain of f , then:

$$\lim_{x \rightarrow a} f(x) = f(a).$$

© Week1 Slides, Page 154

@ CLP-1 by Joel, Andrew, Elyse

$$\lim_{x \rightarrow 5} (x^2 - 4x + 3) \quad \lim_{x \rightarrow 1} (x^7 - 2x^5 + 1)^{35} \quad \lim_{x \rightarrow 2} x(x-1)(x+1) \quad \text{(a)} \lim_{x \rightarrow 3} \left(\frac{x^2 - 9}{x + 3} \right) \quad \lim_{x \rightarrow 0} \left(\frac{6x - 9}{x^3 - 12x + 3} \right)$$

- **Factorising**

$$\lim_{x \rightarrow 2} \left(\frac{x-2}{x^2-4} \right) \quad \lim_{x \rightarrow 4} \frac{x^2-4x}{x^2-16} \quad \lim_{x \rightarrow 1} \frac{x^3-x^2}{x-1} \quad \lim_{x \rightarrow 2} \frac{x^2+x-6}{x-2} \quad \text{(b)} \lim_{x \rightarrow -3} \left(\frac{x^2-9}{x+3} \right)$$

- Anton – Chapter 1.2 – Exercises 3-14
- Anton – Chapter 1.2 – Exercise 40 [This is how exam questions should be]
- Look out for questions that look like “Limits @ Infinity”

$$\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^2 + x - 6} \quad \lim_{x \rightarrow 2} \frac{2x^2 - 5x + 2}{5x^2 - 7x - 6}$$

2. Rationalizing the Denominator

$$\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3} \quad \lim_{y \rightarrow 4} \frac{4-y}{2-\sqrt{y}} \quad \lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1}-1}$$

3. Conjugate

- Anton – Chapter 1.2 – Exercises 37,38

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x} \quad \lim_{x \rightarrow 0} \frac{\sqrt{x^2+4}-2}{x}$$

4. 1 Sided Limits

- PieceWise Functions
- Anton – Chapter 1.2 – Exercises 15-28

$$\lim_{x \rightarrow 0} \frac{x}{|x|} \quad \lim_{x \rightarrow 0} \frac{1}{|2-x|} \quad \lim_{x \rightarrow 3} \frac{x}{x-3} \quad \lim_{x \rightarrow 2} \frac{x}{x^2-4} \quad \text{(c)} \lim_{x \rightarrow 6} \left(\frac{x+6}{x^2-36} \right) \quad \lim_{x \rightarrow 4} \frac{3-x}{x^2-2x-8}$$

- Notice the difference between (a), (b), (c)

5. $\lim_{x \rightarrow \infty}$

• Type I

$$\lim_{x \rightarrow +\infty} \left(\frac{3x+5}{6x-8} \right) = \lim_{x \rightarrow +\infty} \frac{3x}{6x} = \frac{3}{6}$$

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 5x + 2}{5x^2 - 7x - 6} = \lim_{x \rightarrow +\infty} \frac{2x^2}{5x^2} = \frac{2}{5}$$

$$\lim_{x \rightarrow +\infty} \left(\frac{6-x^3}{7t^3+3} \right) = \lim_{x \rightarrow +\infty} \frac{-x^3}{7x^3} = \frac{-1}{7}$$

$$\lim_{x \rightarrow -\infty} \frac{x+4x^3}{1-x^2+7x^3} = \lim_{x \rightarrow -\infty} \frac{4x^3}{7x^3} = \frac{-4}{-7}$$

$$\lim_{x \rightarrow +\infty} \sqrt[3]{\frac{3x+5}{6x-8}} = \sqrt[3]{\lim_{x \rightarrow +\infty} \frac{3x}{6x}} = \sqrt[3]{\frac{3}{6}}$$

$$\lim_{x \rightarrow +\infty} \sqrt[3]{\frac{3x^7 - 4x^5}{2x^7 + 1}} = \sqrt[3]{\lim_{x \rightarrow +\infty} \frac{3x^7}{2x^7}} = \sqrt[3]{\frac{3}{2}}$$

• Type II

$$\lim_{x \rightarrow +\infty} \frac{4x^2 - x}{2x^3 - 5} = \lim_{x \rightarrow +\infty} \frac{4x^2}{2x^3} = \lim_{x \rightarrow +\infty} \frac{4}{2x} = \lim_{x \rightarrow +\infty} \frac{1}{2x} = \lim_{x \rightarrow +\infty} \frac{1}{x} = \frac{1}{\infty} = 0^+$$

$$\lim_{x \rightarrow -\infty} \frac{3}{x+4} = \lim_{x \rightarrow -\infty} \frac{1}{x} = \frac{1}{-\infty} = 0^- \quad \Rightarrow \text{approaching 0, yes, but from the bottom-half of y-axis}$$

$$\lim_{x \rightarrow +\infty} \frac{1}{x-12} = \lim_{x \rightarrow +\infty} \frac{1}{x} = \frac{1}{+\infty} = 0^+$$

$$\lim_{x \rightarrow -\infty} \frac{x-2}{x^2+2x+1} = \lim_{x \rightarrow -\infty} \frac{x}{x^2} = \lim_{x \rightarrow -\infty} \frac{1}{x} = 0^-$$

• SideNote: $\lim_{-\infty}$

$$\lim_{x \rightarrow -\infty} \frac{x}{x^2} = \lim_{x \rightarrow -\infty} \frac{1}{x} \cdot \frac{x}{x} \quad \Rightarrow \quad \lim_{x \rightarrow -\infty} \frac{1}{x} \cdot \frac{-ve}{-ve} \quad \Rightarrow \quad \lim_{x \rightarrow -\infty} \frac{1}{x} \cdot 1 = \lim_{x \rightarrow -\infty} \frac{1}{x} = \frac{1}{-\infty} = 0^-$$

• Type III

$$\lim_{x \rightarrow +\infty} \frac{7-6x^5}{x+3} = \lim_{x \rightarrow +\infty} \frac{-6x^5}{x} = \lim_{x \rightarrow +\infty} (-6) \cdot \frac{x^4}{1} = (-6) \cdot \frac{(+\infty)^4}{1} = (-6) \cdot \frac{\infty}{1} = -\infty$$

$$\lim_{x \rightarrow -\infty} \frac{7-6x^5}{x+3} = \lim_{x \rightarrow -\infty} \frac{-6x^5}{x} = \lim_{x \rightarrow -\infty} (-6) \cdot \frac{x^4}{1} = (-6) \cdot \frac{(-\infty)^4}{1} = (-6) \cdot \frac{\infty}{1} = -\infty$$

$$\lim_{x \rightarrow -\infty} \frac{7-6x^4}{x+3} = \lim_{x \rightarrow -\infty} \frac{-6x^4}{x} = \lim_{x \rightarrow -\infty} (-6) \cdot \frac{x^3}{1} = (-6) \cdot \frac{(-\infty)^3}{1} = (-6) \cdot \frac{-\infty}{1} = +\infty$$

• Anton – Chapter 1.3 – Page 93: “A Quick method for finding limits of ...”

• Anton – Chapter 1.3 – Exercises 09-24

• Exercises

$$\lim_{x \rightarrow -\infty} \frac{3}{x+4} \quad \lim_{x \rightarrow +\infty} \frac{1}{x-12} \quad \lim_{x \rightarrow -\infty} \frac{x-2}{x^2+2x+1} \quad \lim_{x \rightarrow +\infty} \frac{6-x^3}{7x^3+3} \quad \lim_{x \rightarrow -\infty} \frac{x+4x^3}{1-x^2+7x^3}$$

5. $\lim_{x \rightarrow \infty}$

• Type IV – Divide by $|x|$

- Anton – Chapter 1.3 – Page 94: “Limits Involving Radicals”

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x \rightarrow +\infty} \frac{\left(\frac{\sqrt{x^2+2}}{|x|}\right)}{\left(\frac{3x-6}{|x|}\right)} = \lim_{x \rightarrow +\infty} \frac{\left(\frac{\sqrt{x^2+2}}{\sqrt{x^2}}\right)}{\left(\frac{3x-6}{+x}\right)} \quad \because |x| = \sqrt{x^2} \quad \text{and} \quad \because |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+2}}{3x-6} = \lim_{x \rightarrow -\infty} \frac{\left(\frac{\sqrt{x^2+2}}{|x|}\right)}{\left(\frac{3x-6}{|x|}\right)} = \lim_{x \rightarrow -\infty} \frac{\left(\frac{\sqrt{x^2+2}}{\sqrt{x^2}}\right)}{\left(\frac{3x-6}{-x}\right)} \quad \because |x| = \sqrt{x^2} \quad \text{and} \quad \because |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

- Anton – Chapter 1.3 – Exercises 25-30
- Quick Method

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2}-2}{x+3} = \lim_{x \rightarrow -\infty} \frac{\sqrt{5x^2}}{x} = \frac{\sqrt{5}}{-1} \qquad \lim_{x \rightarrow +\infty} \frac{\sqrt{5x^2}-2}{x+3} = \lim_{x \rightarrow +\infty} \frac{\sqrt{5x^2}}{x} = \frac{\sqrt{5}}{+1}$$

$$\lim_{x \rightarrow -\infty} \frac{2-x}{\sqrt{7+6x^2}} = \lim_{x \rightarrow -\infty} \frac{-x}{\sqrt{6x^2}} = \frac{-(-1)}{\sqrt{6}}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4}+x}{x^2-8} = \lim_{x \rightarrow -\infty} \frac{\sqrt{3x^4}}{x^2} = \frac{\sqrt{3}}{1}$$

• Type V: Multiply by Conjugate

$$\lim_{x \rightarrow +\infty} \sqrt{x^6+5} - x^3 \qquad \lim_{x \rightarrow +\infty} \sqrt{x^6+5x^3} - x^3$$

- Anton – Chapter 1.3 – Exercises 31-32 [Conjugate]

$$\lim_{x \rightarrow +\infty} \sqrt{x^2+3} - x \qquad \lim_{x \rightarrow +\infty} \sqrt{x^2-3x} - x$$

• Type VI

- Anton – Chapter 1.3 – Page 95: “End behavior of Trigonometric, Exponential, and Logarithmic Funcs”
- Anton – Chapter 1.3 – Exercises 33-36 – Divide by e^x

$$\lim_{x \rightarrow -\infty} \frac{1-e^x}{1+e^x}$$

- Anton – Chapter 1.3 – Exercises 37-38 – $\ln()$

$$\lim_{x \rightarrow +\infty} \ln\left(\frac{2}{x^2}\right) \qquad \lim_{x \rightarrow 0^+} \ln\left(\frac{2}{x^2}\right)$$

6. Squeeze Theorem / SandWich Theorem

$$\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1$$

$$\lim_{h \rightarrow 0} \cos\left(\frac{1}{h}\right) = \text{range}[-1, 1] = \text{D.N.E.}$$

$$\lim_{h \rightarrow 0} h \cdot \cos\left(\frac{1}{h}\right) = 0$$

$$\lim_{h \rightarrow 0} h \cdot \sin\left(\frac{1}{h}\right) = 0$$

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\cos(h)-1}{h} &= \lim_{h \rightarrow 0} \frac{\cos(h)-1}{h} \cdot \frac{\cos(h)+1}{\cos(h)+1} \\ &= \lim_{h \rightarrow 0} \frac{\cos^2(h)-1^2}{h} \cdot \frac{1}{\cos(h)+1} \\ &= \lim_{h \rightarrow 0} \frac{1-\sin^2(h)-1^2}{h} \cdot \frac{1}{\cos(h)+1} \\ &= \lim_{h \rightarrow 0} \frac{-\sin^2(h)}{h} \cdot \frac{1}{\cos(h)+1} \\ &= \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \cdot -\frac{\sin(h)}{\cos(h)+1} \end{aligned}$$